

Snaky in 21 Maker moves

OpenAI

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Abstract

We prove that Maker can achieve the Snaky hexomino within 21 actual Maker moves against arbitrary legal Breaker play on the initially empty infinite square board. The same bound holds on a 17×17 square; in fact, Maker can confine its claims to a fixed 251-cell board.

1 Introduction

Polyomino achievement games ask whether one player can obtain a fixed shape despite an opponent’s attempts to obstruct it. In the weak achievement game, usually called the Maker–Breaker game, only Maker seeks to complete the shape. The opponent need not make a competing copy. Snaky is the six-cell target

$$S = \{(0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (4, 1)\} \subset \mathbb{Z}^2. \quad (1)$$

Let $\mathcal{G}$ be the eight 2×2 signed permutation matrices: each row and column has one nonzero entry, equal to 1 or -1 . An allowed copy is $t + R(S)$ with $t \in \mathbb{Z}^2$ and $R \in \mathcal{G}$. Thus reflections and quarter turns are permitted, and scaling is not. Figure 1 shows the target in the coordinates of (1).

The board is all of $\mathbb{Z}^2$, initially empty. Maker moves first, and thereafter Maker and Breaker alternate, each claiming one previously unclaimed cell. Ownership is permanent. Maker wins immediately upon owning every cell of an allowed copy, and may own additional cells. There are no initially owned cells or additional handicap moves. A strategy chooses a legal cell from the complete finite history.

Theorem 1. *In this game, there is one globally legal Maker policy that completes an allowed copy of S within at most 21 actual Maker claims against every legal continuation of Breaker. Equivalently, extend the policy by fresh claims after an earlier win: its Maker set after claim 21 contains a target whenever the first 20 Breaker replies are legal.*

The bound counts Maker’s actual claims, including any replacement moves used in the proof. It corresponds to a win no later than the 41st individual turn of the alternating game. Corollary 6 also gives the same bound on a finite 17×17 board. We do not assert optimal move counts or board sizes, or a result for the strong game in which both players seek a copy.

Earlier results and methods. Sieben attributes the introduction of polyomino achievement games to Frank Harary [6, Section 1]. His account records handicap-two strategies of Harary, Harborth and Seemann and the nonexistence of a pairing defense for Snaky, and proves victory in 41 dimensions [6, Section 4 and Proposition 4.1]. Csernenszky, Martin and Pluhár later gave a computer-free proof of the nonexistence of a pairing defense [2, Theorem 10]. This rules out a class of Breaker strategies, but does not by itself give Maker a winning strategy.

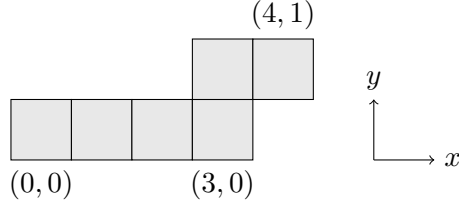


Figure 1: The six cells of Snaky. A cell is indexed by the coordinates of its lower left corner. Every integral translation and signed coordinate permutation of this shape is an allowed target.

Halupczok and Schlage-Puchta proved that Snaky wins in three dimensions, loses on an 8×8 board, and has planar handicap number at most one [3, Theorem 1]. A handicap of one allows Maker two stones on the opening turn, before Breaker’s first reply. Ito and Miyagawa also presented a handicap-one construction [4]. Sieben subsequently gave a finite proof sequence for that handicap and described the ordinary planar problem as undecided at the time [7, Section 1 and Appendix B].

Theorem 1 treats the empty planar board without a handicap. We give three complete constructions, with bounds of 21, 25, and 35 Maker moves. They share a composition rule but retain different information and different openings. The 21-move certificate ends directly with no required Maker stones. The 25-move construction removes already-owned cells from each Breaker-free region and uses required cells as placement anchors. The 35-move construction supplies four explicit conditional statements at arbitrary positions and an opening that reaches one of them. We keep these methods explicit because their content is not determined by the empty-board move bounds alone.

Our method belongs to the proof-tree approach to weak achievement studied by Sieben [7, Sections 2–3]. His finite situations and territory-intersection conditions encode ways of combining continuations at Breaker’s turn. Our conditional positions are stated at Maker’s turn, with a separate Breaker-turn consequence for openings in which the pivot is already owned. The formulation explicitly keeps track of arbitrary previous Maker and Breaker ownership and counts actual future Maker moves. We prove its soundness in Section 2. The perpendicular attacks discussed by Halupczok and Schlage-Puchta [3, Section 3] also illuminate why a line of four Maker stones can exert substantial pressure. Section 5 gives a separate, fully stated local result of this kind.

Shaik, Mayer-Eichberger, van de Pol and Saffidine studied bounded-depth QBF encodings of finite Maker–Breaker games, including Snaky on a 5×5 board [5, Sections 2.1 and 6]. Boucher and Villemaire study QBF encodings for finite strong achievement games on ordinary and toroidal boards [1, Sections 2.2, 3.3–3.4, and 4]. Here the finite object is a proof of sufficient conditions for the infinite weak game. Breaker is free to play outside every displayed region, and the proof covers those replies without a board cutoff.

Proof overview. A conditional winning position is described by three finite data: a set A that Maker must own, a set T that must contain no Breaker cell, and a bound h on further Maker moves. Starting with the six immediate completions of S , we combine such conditions as follows. Maker claims a chosen cell. The required cells of every child condition then belong to Maker; moreover, the intersection of the child envelopes belongs to Maker. Consequently one new Breaker cell cannot lie in every child envelope, and at least one continuation remains available.

Section 3 specifies 728 numbered conditions, including the six bases, by exact integer coordinates and backward references. The final one has $A = \emptyset$, an envelope of 251 cells, and height 21. Section 4 turns the recursive proof into one policy chosen before Breaker’s play. The independent four-in-a-row result follows in Section 5. Section 6 distinguishes the mathematical induction, exact data identities,

and independent finite calculations. Appendix A proves the reduced-envelope construction and its five-class opening. Appendix B gives the complete 35-move construction, its unbounded opening, and its conditional tactics. Appendix C describes the finite reconstruction supplement. Appendices D–F print the three literal certificates, and Appendix G explains the standalone verification programs.

2 Conditional winning positions

The purpose of this section is to justify a finite rule that remains valid when Breaker may play anywhere on the infinite board. At a finite position let $M, B \subset \mathbb{Z}^2$ be the finite disjoint sets currently owned by Maker and Breaker. Their complete contents are retained throughout the argument.

Definition 2. Let $A \subseteq T \subset \mathbb{Z}^2$ be finite and let $h \geq 1$ be an integer. The triple (A, T, h) is a *valid claim* if, at every position with Maker to move and with

$$A \subseteq M, \quad B \cap T = \emptyset,$$

Maker has already completed a target or has a strategy that completes one within h further actual Maker moves. We call A the required set, T the envelope, and h the height.

This definition imposes no restriction on additional Maker cells, on Breaker cells outside T , or on how the position was reached. In particular, it applies to arbitrary finite disjoint ownership, without requiring a prescribed difference between $|M|$ and $|B|$. Only the starting turn is specified. A *placement* is an affine map $F(x) = Rx + t$ with $R \in \mathcal{G}$ and $t \in \mathbb{Z}^2$. A valid claim remains valid under the simultaneous replacement of A, T by $F(A), F(T)$, with the same height: the bijection F transports a legal strategy, preserves disjointness and targets, and does not change the number of moves.

The use of finite sufficient conditions and alternative continuations has a close antecedent in Sieben’s proof trees and situations [7, Sections 2–3]. His situations consist of a Maker core and a neighborhood without Breaker stones and are stated at Breaker’s turn; the proof trees use an empty intersection of continuation territories. The rule below works with arbitrary finite prior ownership and counts every actual fresh Maker claim. Its quantitative form applies to all three constructions in this article.

Lemma 3 (Combination rule). *Let (A_i, T_i, h_i) , $1 \leq i \leq r$, be valid claims, where $1 \leq r < \infty$, and let $p \in \mathbb{Z}^2$. Define*

$$T = \{p\} \cup \bigcup_{i=1}^r T_i, \quad A = \left(\bigcup_{i=1}^r A_i \cup \bigcap_{i=1}^r T_i \right) \setminus \{p\}, \quad h = 1 + \max_i h_i. \quad (2)$$

Then (A, T, h) is valid. Also, at a position with Breaker to move, if $A \cup \{p\} \subseteq M$ and $B \cap T = \emptyset$, then Maker has already won or can win within $h - 1$ further Maker moves.

Proof. Every A_i is contained in T_i , so $A \subseteq T$. Suppose first that Maker is to move, $A \subseteq M$, $B \cap T = \emptyset$, and no target is yet complete. Since $p \in T$, Breaker does not own p . If p is free, Maker claims it. If $p \in M$, Maker instead claims any free cell. Such a cell exists because the board is infinite and $M \cup B$ is finite. This replacement is one real move. In both cases the resulting Maker set M' contains

$$\bigcup_i A_i \cup \bigcap_i T_i \subseteq A \cup \{p\} \subseteq M'. \quad (3)$$

If this move completes a target, stop immediately. Otherwise let b be the next legal Breaker cell. As $b \notin M'$, (3) implies $b \notin \bigcap_i T_i$. Choose an index i with $b \notin T_i$. The older Breaker cells also avoid T_i , since $T_i \subseteq T$. Thus, at the next Maker turn,

$$A_i \subseteq M', \quad (B \cup \{b\}) \cap T_i = \emptyset.$$

The child claim gives a win within h_i more Maker moves. Including the move already spent gives at most $1 + h_i \leq h$.

For the Breaker-turn assertion, the inclusions in (3) already hold with M in place of M' . After Breaker's next move the same surviving child applies, without spending the parent Maker move. Its bound is at most $\max_i h_i = h - 1$. $\square$

The replacement move cannot invalidate this argument: it only adds Maker ownership, and its actual ensuing Breaker reply is included in the proof. It must not be omitted from the count. A reply outside T misses every child envelope, so distant play requires no separate case analysis. The Breaker-turn assertion also shows that p need not have been Maker's most recent claim.

Lemma 4 (Six bases). *Write the points of S in the order in (1) as $s_0, \dots, s_5$. The claims*

$$(A_j, T_j, h_j) = (S \setminus \{s_j\}, S, 1), \quad 0 \leq j < 6, \quad (4)$$

are valid.

Proof. If Maker owns s_j , it already owns all of S . Otherwise s_j is free because Breaker owns none of S . Claiming it completes the target in one move. $\square$

Every application of Lemma 3 may use placed versions of earlier claims. We will describe a finite collection of such applications and calculate its last required set and height. The semantic proof is already complete: once that required set is empty, the final height bounds play from the empty board.

3 The certificate and its final condition

This section specifies the finite applications of the combination rule. We use the word *card* for a numbered claim in this 21-move certificate. Its numbering is independent of the 25- and 35-move tables in Appendices A and B. Numbered references allow a previously proved claim to be reused; an expression written in parentheses is simply an additional, unnumbered application of the same rule. The complete literal data are in Appendix D.

3.1 Coordinates, placements, and expressions

The coordinate alphabet is

0123456789ABCDEFGF.

A two-character word $\mathbf{XY}$ denotes the point (x, y) whose coordinates are the positions of $\mathbf{X}$ and $\mathbf{Y}$ in this alphabet, counting from zero. For example, $\mathbf{88}$ means $(8, 8)$, $\mathbf{7A}$ means $(7, 10)$, and $\mathbf{GG}$ means $(16, 16)$. The encoded pivots and translation vectors have nonnegative coordinates; the sets produced after signed transformations may contain negative coordinates.

A reference $\mathbf{j:sUV}$ means: take numbered card j , apply the signed coordinate permutation R_s , and then translate by the point encoded by $\mathbf{UV}$. To compute R_s , negate the first coordinate if the

bit of weight 2 is set, negate the second if the bit of weight 4 is set, and *then* interchange the coordinates if the bit of weight 1 is set. The resulting maps are

$$\begin{array}{c|cccc} s & 0 & 1 & 2 & 3 \\ \hline R_s(x, y) & (x, y) & (y, x) & (-x, y) & (y, -x) \end{array} \quad \begin{array}{c|cccc} s & 4 & 5 & 6 & 7 \\ \hline R_s(x, y) & (x, -y) & (-y, x) & (-x, -y) & (-y, -x). \end{array} \quad (5)$$

These are precisely the eight matrices in $\mathcal{G}$. A bare reference j uses the identity placement. References change neither the child's height nor the meaning of its winning condition.

Every data line consists of its decimal card number j , a pivot coordinate, and a nonempty ordered list of children. Each child is either a reference or an expression of the form

$$(\mathbf{XY} \text{ child child } \dots).$$

Such an expression combines its children at the pivot $\mathbf{XY}$ using (2). All coordinates inside parentheses remain in the coordinate frame of the line. They are not offsets from the enclosing pivot. When the entire card is later placed by an affine map, that map transports every pivot and set in its proof.

Cards $0, \dots, 5$ are the six bases (4). The data specify cards $6, \dots, 727$ in order. Every numbered reference in a line, including a reference inside parentheses, is to a smaller card number. We evaluate innermost expressions first, using (2) for the sets and height. Consequently structural induction on each expression, followed by induction on the line number, proves that every resulting card is a valid claim. The finite calculation determines which sufficient conditions this particular list produces.

3.2 Two elementary cards

The first two lines illustrate both the geometry and the encoding:

```
6 14 4:100 5:101
7 05 6:001 6:405
```

For card 6, the two bases are first transposed; the second is also translated by $(0, 1)$. Direct substitution gives

$$A_6 = \{(0, y) : 0 \leq y \leq 4\}, \quad T_6 = A_6 \cup \{(1, 3), (1, 4), (1, 5)\}, \quad h_6 = 2. \quad (6)$$

Claiming $(1, 4)$ prepares two copies, completed respectively at $(1, 3)$ and $(1, 5)$. Unless a copy is already complete, Breaker can occupy at most one of these cells. Figure 2 shows the two alternatives.

For card 7, the two copies of A_6 both occupy the column $x = 0$, $1 \leq y \leq 5$. Their additional envelope cells have heights 4, 5, 6 and 0, 1, 2, respectively. These triples are disjoint, so the intersection of the envelopes is precisely the common column. Deleting the pivot $(0, 5)$ therefore gives

$$\begin{aligned} A_7 &= \{(0, y) : 1 \leq y \leq 4\}, & h_7 &= 3, \\ T_7 &= \{(0, y) : 1 \leq y \leq 5\} \cup \{(1, y) : y \in \{0, 1, 2, 4, 5, 6\}\}. \end{aligned}$$

The general certificate repeats this mechanism with longer conditional continuations and cells that need not lie near the current line of Maker stones.

3.3 The final card

Proposition 5. *The data in Appendix D have exactly 722 numbered lines, with indices $6, \dots, 727$. Every line has a nonempty child list, every reference uses an allowed placement of an earlier card,*

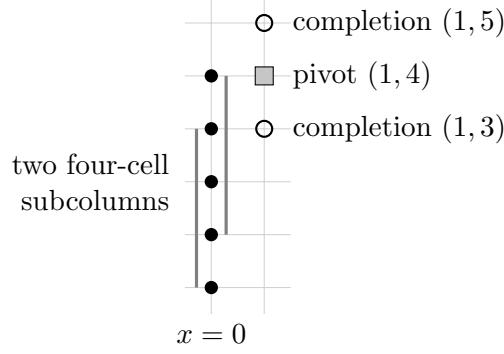


Figure 2: Card 6: dots are required Maker cells, the square is the next pivot, and circles mark the two possible completions. The vertical guides indicate the two four-cell bodies. The envelope contains no Breaker cell; any of its additional cells may already belong to Maker.

and every parenthesized expression is a finite nonempty combination. Together with the six bases, they produce 728 valid claims. Their heights are at most 21, and the last card has

$$p_{727} = (8, 8), \quad A_{727} = \emptyset, \quad |T_{727}| = 251, \quad h_{727} = 21. \quad (7)$$

Moreover, $T_{727} \subseteq \{0, \dots, 16\}^2$.

Proof. The syntactic and set assertions are a finite calculation from (4), (2), and (5), using the full displayed data. The verifier in Appendix G reconstructs every numbered card. Its arrays initially contain the six bases. Before line j is processed, their entries at indices below j are exactly the sets and heights specified by the preceding lines. A reference retrieves an earlier entry and applies its stated placement. Recursive evaluation applies (2) at every parenthesized node and at the root of the line. The nonempty-child and backward-reference tests make every union, intersection, maximum, and lookup well-defined. This proves the invariant for the next line and identifies the computed sets with the mathematically defined ones. The final tests give (7), maximum height 21, and the intermediate values in Table 1. The separate reconstruction `verification/supporting/independent_evaluator.py` also computes all sets and heights and checks the successive intersections. Validity then follows from Lemmas 3 and 4, by the structural induction just described. $\square$

For clarity, Table 1 records how the final empty requirement arises. The last line has 32 placed children, grouped by their six card numbers. Each placed required set is exactly $\{(8, 8)\}$. The last column is the number of cells other than $(8, 8)$ in the intersection of the envelopes from that group and all preceding groups. The full placements occur in the last line of the certificate.

In particular, the intersection of all child envelopes is exactly $\{(8, 8)\}$: it contains that point because every child requirement does, and the last table entry removes every other point. The union of the child requirements is the same singleton. Equation (2) therefore gives $A_{727} = \emptyset$. The maximum child height is 20, giving the bound 21.

The data also illustrate how play may leave an immediately adjacent line. After Maker plays $(8, 8)$ and Breaker replies $(7, 8)$, the placed child 708:100 is available. Its pivot is $(8, 7)$. If Breaker next takes $(8, 6)$, the child 684:011 of card 708, still under the outer transposition, is available and has pivot $(11, 8)$. These are legal illustrative choices, not imposed responses by Breaker. Every other reply is covered by the same intersection rule.

Card j	A_j	$ T_j $	h_j	Placements	Remaining cells
648	$\{(4, 5)\}$	87	15	4	35
708	$\{(8, 8)\}$	225	20	4	31
712	$\{(5, 5)\}$	96	15	8	21
713	$\{(5, 5)\}$	98	15	4	12
725	$\{(7, 7)\}$	167	20	8	4
726	$\{(7, 7)\}$	169	20	4	0

Table 1: The six groups of children in card 727. All their requirements become the pivot. Their common envelope has no other cell, so the combination rule leaves an empty required set.

4 One strategy against every continuation

We now turn the final card into a single strategy. Making its choices definite also clarifies that the height counts actual Maker moves.

Proof of Theorem 1. Fix an enumeration of $\mathbb{Z}^2$: order cells by increasing $\max(|x|, |y|)$, then lexicographically within each finite level. Retain the printed order of the children in every combination expression. These choices are fixed before any play. Initially the active claim is card 727 in its identity placement. Its hypotheses hold on the empty board by (7).

At an active combination with current placement F , claim its placed pivot $F(p)$ if free. If Maker already owns $F(p)$, claim the first free cell in the fixed enumeration instead. The invariant of Definition 2 guarantees that Breaker does not own the pivot, and finiteness of ownership guarantees a free replacement. Stop if a target is complete. Otherwise, after Breaker’s next legal reply, choose the first child in the printed list whose placed envelope avoids that reply. Lemma 3 proves that such a child exists and that all its requirements and its full envelope satisfy the invariant, including the older Breaker cells. For a referenced card whose reference placement is G , the new placement is $F \circ G$; for an inline expression, retain F and use that expression.

At a placed base card, either the target is already present or its one missing cell is free. In the latter case claim it and win. Thus a nonterminal step spends one actual Maker move and passes, after one actual Breaker reply, to a child of strictly smaller height. The bases have height 1. Starting at height 21, this procedure wins within 21 Maker moves.

The active expression and placement are determined by the complete history: replay the prescribed pivot or replacement choices and the first-surviving-child choices from card 727. On a history inconsistent with an earlier prescribed Maker choice, use the first free cell of the enumeration. Use the same fallback on an already-won history if one chooses to continue it formally. This defines a globally legal policy on all finite legal histories, including histories not reached from the empty board under the policy. No choice of the policy depends on future Breaker moves. The preceding induction applies to every legal continuation from the empty board and proves the theorem. $\square$

Only replies before victory are used in this argument. If the first win occurs on Maker move $m \leq 21$, there have been precisely $m - 1$ Breaker moves. In particular, no reply after Maker’s winning move is assumed or required.

One can express the same quantifier order using a fixed endpoint. Extend the policy beyond a completed target by the first-free-cell rule. For every sequence of proposed Breaker cells whose first 20 entries are legal along this extended policy, the Maker set immediately after move 21 contains an allowed copy of S . An earlier completed copy persists because ownership is permanent. This formulation is $\exists\sigma \forall\beta$, with σ chosen in the proof and β the sequence of Breaker replies; it makes no

assumption about a twenty-first Breaker reply. Stopping at the first target recovers the ordinary game.

Corollary 6 (A finite board). *Maker wins the weak Maker–Breaker game within 21 actual Maker claims on the initially empty 251-cell board T_{727} , and also on the initially empty 17×17 board $\{0, \dots, 16\}^2$. In each game, Maker moves first, the players alternate claiming one previously unclaimed cell, and the targets are the allowed copies of S contained in the board.*

Proof. Evaluation of the literal certificate gives

$$T_{727} \subseteq \{0, \dots, 16\}^2,$$

in addition to the endpoint data in (7). Every placed child envelope is contained in its parent envelope, and every pivot and base target lies in its own envelope. Thus all prescribed pivots and base completions lie in T_{727} . Use that strategy, replacing an already-owned pivot by the first free cell in a fixed ordering of T_{727} . Before Maker claim $m \leq 21$, at most $2(m - 1) \leq 40$ cells have been occupied, so such a replacement is always available on either board.

After a pivot or replacement claim, Maker owns every child requirement and the intersection of the child envelopes. Every legal Breaker reply therefore leaves a surviving child envelope; nesting also keeps the older Breaker cells outside that envelope. This includes replies outside T_{727} on the square board. The same height induction now gives victory within 21 actual Maker claims, counting each replacement. Play stops as soon as a target is complete; no Breaker reply after victory is used. $\square$

5 Four stones in a row

The certificate establishes the empty-board theorem without assuming that nearby Breaker cells stay sparse. A separate local statement explains some of the geometry behind its threats. The perpendicular attack below is related to the four-in-a-row tactics of Halupczok and Schlage-Puchta [3, Section 3, pp. 13–15]; their planar row extensions also appear in Sections 4.4–4.6 of that paper. We give a complete proof for the precise finite-neighborhood hypothesis used here.

Proposition 7. *Suppose it is Maker’s turn at an arbitrary finite position, Maker owns the four cells*

$$\{(0, 0), (1, 0), (2, 0), (3, 0)\},$$

and Breaker owns at most three cells in

$$Q = ([-3, 6] \times [-4, 4]) \cap \mathbb{Z}^2.$$

Then Maker has already won or can force a win. Additional Maker cells and arbitrary Breaker cells outside Q are allowed.

Call a set *clean* if it contains no Breaker cell. Throughout this section, an instruction to claim an already-owned Maker cell means to claim any free cell instead, unless a target has already been completed. Such a replacement is a real move and allows exactly one new Breaker reply. Extra Maker cells therefore cause no difficulty in any of the arguments below.

When a move is said to force Breaker to a cell q , the precise meaning is that q completes an allowed target on Maker’s next move. If q is already Maker’s, the target is already complete. Otherwise q is free, and Breaker must claim it or allow that immediate completion. Every forced continuation is considered only when Maker has not already won or obtained this next-move win.

5.1 A five-cell fork

Suppose Maker owns the horizontal five $\{(r, 0), \dots, (r + 4, 0)\}$. For an endpoint $e \in \{r, r + 4\}$ and a sign $\varepsilon \in \{-1, 1\}$, put

$$H(e, \varepsilon) = \{(e - 1, \varepsilon), (e, \varepsilon), (e + 1, \varepsilon)\}. \quad (8)$$

If this triple is clean at Maker's turn, claiming its middle cell (e, ε) creates two one-cell completions, at its other two cells. The two targets use respectively the first and last four-cell subrows of the five; they are reflected or translated copies of S . Breaker can occupy at most one completion, so Maker wins on the next move unless it has already won. This is the horizontal version of Figure 2.

The four triples in (8) are pairwise disjoint. Consequently, if Maker has just created the five and at least two of these triples are clean *before* Breaker's response, at least one remains clean afterwards, and the fork wins. The same assertion holds after interchanging the coordinates. For a vertical five we use the notation

$$V(c, d) = \{(c, d - 1), (c, d), (c, d + 1)\},$$

where $c \in \{-1, 1\}$ and d is an endpoint height.

5.2 The perpendicular endpoint attack

We first establish a local tactic at the right endpoint of a row. Assume Maker owns the four cells

$$(-3, 0), \quad (-2, 0), \quad (-1, 0), \quad (0, 0),$$

and the set

$$Z = \{(x, y) : x \in \{-1, 0, 1\}, 1 \leq |y| \leq 4\} \quad (9)$$

is clean. Other ownership is arbitrary. Maker claims $(0, 1)$, forcing $(1, 1)$, and then $(0, -1)$, forcing $(1, -1)$. Indeed, the original horizontal body with the respective pair of cells above or below its endpoint is an allowed Snaky. If Breaker declines either forced reply, Maker completes that copy on its next turn.

After those two replies, Maker claims $(0, 2)$. Denote the new Breaker reply by b , and set

$$E = \{(0, -2)\} \cup V(-1, -2) \cup V(-1, 2).$$

Figure 3 separates the two possible continuations after this reply. If $b \notin E$, Maker claims $(0, -2)$. It now owns the vertical five from height -2 to height 2 , and both $V(-1, -2)$ and $V(-1, 2)$ are clean before the ensuing Breaker reply. They are disjoint, and neither of the two earlier forced Breaker cells is in either triple. The five-cell fork therefore wins.

If $b \in E$, Maker instead claims $(0, 3)$, creating the vertical five from height -1 to height 3 . The triple $V(1, 3)$ is clean. The triple $V(-1, 3)$ is also clean unless

$$b = (-1, 2) \quad \text{or} \quad b = (-1, 3).$$

If both are clean, the same two-triple argument wins. In either exceptional case there is instead an immediate completion at $q = (-1, -1)$, because Maker already owns the other five cells of

$$\{(0, 3), (0, 2), (0, 1), (0, 0), (-1, 0), (-1, -1)\} = \{(-y, 3 - x) : (x, y) \in S\}. \quad (10)$$

The cell q lies in the initially clean zone, and differs from b and from the two earlier forced replies. It is thus free unless Maker already owns it, in which case the target is complete. It also lies outside

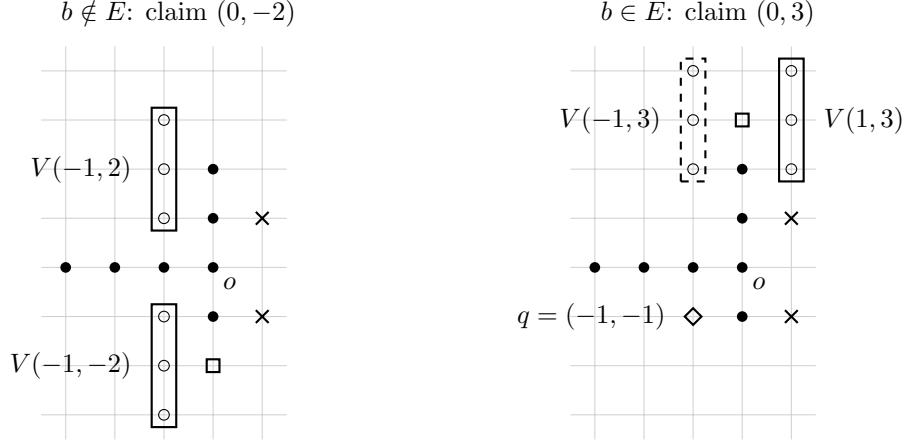


Figure 3: Alternative endpoint continuations after the reply b , which is not drawn; here $o = (0, 0)$. Dots show guaranteed Maker cells, crosses the two earlier forced Breaker cells, and each square the prescribed next claim. Solid outlines enclose clean triples, whose cells may already be Maker-owned. On the left, their union with the square is E . On the right, the dashed triple is clean except when $b = (-1, 2)$ or $b = (-1, 3)$; in those cases, claiming $(0, 3)$ leaves the diamond q as an immediate completion while $V(1, 3)$ stays clean. The panels are conditional alternatives, not consecutive moves.

$V(1, 3) = \{(1, 2), (1, 3), (1, 4)\}$. Breaker must either claim q , leaving that clean triple for the five-cell fork, or leave q as an immediate winning move. This completes the endpoint tactic.

Every prescribed target or threat cell in this tactic lies in $\{-3, -2, -1, 0\} \times \{0\}$ or in Z . Fresh replacement moves may be elsewhere on the board. In particular, arbitrary Breaker moves outside the stated zone are included in the preceding cases and cannot defeat the tactic.

5.3 Proof of the four-in-a-row result

Proof of Proposition 7. Assume no target is already complete, and consider the two extensions $(-1, 0)$ and $(4, 0)$ of the given row.

If both extensions belong to Breaker, there is at most one further old Breaker cell in Q . The endpoint tactic has two possible placements: at the left endpoint use $(x, y) \mapsto (-x, y)$, and at the right endpoint use $(x, y) \mapsto (x + 3, y)$. Their off-row zones are respectively

$$\{-1, 0, 1\} \times \{-4, -3, -2, -1, 1, 2, 3, 4\} \quad \text{and} \quad \{2, 3, 4\} \times \{-4, -3, -2, -1, 1, 2, 3, 4\}.$$

They are disjoint subsets of Q , and neither contains the two row blockers. At least one zone is clean. The corresponding endpoint tactic proves the result.

If exactly one extension belongs to Breaker, claim the other. This produces a five in a row. The four endpoint triples are pairwise disjoint and all lie in Q . The old row blocker meets none of them, and at most two further old Breaker cells can meet them. Thus at least two triples are clean before the new Breaker response, and the five-cell fork wins.

It remains to consider the case that neither extension belongs to Breaker. If one extension produces a five with at least two clean endpoint triples, use it and finish as above. Suppose that both choices fail this test. For extension to the left, the four triples are $H(-1, \pm 1), H(3, \pm 1)$; for extension to the right, they are $H(0, \pm 1), H(4, \pm 1)$. Failure of either test requires at least three of its disjoint triples to contain an old Breaker cell. Since at most three old Breaker cells lie in Q , there are exactly three, and each must meet a triple for both tests. A cell with this property lies on

level -1 or 1 and has its first coordinate in $\{-1, 0\}$ or in $\{3, 4\}$. Moreover, their three combinations of *left or right cluster* and *lower or upper level* are distinct; otherwise fewer than three disjoint triples would be met in one of the tests.

The reflections $(x, y) \mapsto (3 - x, y)$ and $(x, y) \mapsto (x, -y)$ preserve both the original four-cell row and Q . They allow us to take the missing cluster-level combination to be the lower right one. The three old Breaker cells then have the form

$$(l_-, -1), \quad (l_+, 1), \quad (r, 1), \quad l_-, l_+ \in \{-1, 0\}, \quad r \in \{3, 4\}. \quad (11)$$

No other old Breaker cell lies in Q . Let a count the true statements $l_- = 0$, $l_+ = 0$, and $r = 4$.

Suppose first that $a \geq 2$. Claim $(-1, 0)$. The resulting five from -1 to 3 has the clean triple $H(3, -1)$. Unless Breaker meets it on its next move, Maker uses that fork. Hence we may assume Breaker's new cell belongs to $H(3, -1) = \{(2, -1), (3, -1), (4, -1)\}$. Claim $(-2, 0)$, which is free or already Maker's: neither an old blocker nor that new cell can occupy it. For the five from -2 to 2 , each of the following is a clean endpoint triple when its indicated condition holds:

triple	condition
$H(-2, -1)$	$l_- = 0$
$H(-2, 1)$	$l_+ = 0$
$H(2, 1)$	$r = 4$.

They are pairwise disjoint and are all missed by the last Breaker move. There are at least $a \geq 2$ such triples before the next response. The five-cell fork wins.

If $a \leq 1$, claim $(4, 0)$ instead. Its clean lower-right triple $H(4, -1)$ forces the next Breaker cell into $\{(3, -1), (4, -1), (5, -1)\}$, or gives an immediate fork win. Claim $(5, 0)$, again free or already Maker's. For the five from 1 to 5 , the clean disjoint triples are

triple	condition
$H(1, -1)$	$l_- = -1$
$H(1, 1)$	$l_+ = -1$
$H(5, 1)$	$r = 3$.

At least $3 - a \geq 2$ conditions hold, and the most recent Breaker cell meets none of these triples. The fork again wins. All extension cells and triples in both cases lie in Q . This exhausts the possibilities and proves the proposition. $\square$

Proposition 7 is a sufficient condition at an arbitrary Maker turn. Its proof does not assume that the four stones were obtained consecutively, and the 21-move theorem does not assume that its three-blocker hypothesis occurs during play. The full certificate supplies the empty-board strategy.

6 Finite verification and its scope

The three certificates describe finite applications of proved rules. Their verification has three separate parts: the game-theoretic soundness of those rules, the exact finite sets produced by the data, and the identity of the distributed files. A checksum addresses only the last part. The mathematical implication from a computed card to every legal continuation is Lemma 3, together with the reduced-envelope bridge in Appendix A.

Construction	Base cards	Numbered nonbase rows	References	Final bound
21 moves	6	722	4,089	21
25 moves	6	2,010	5,862	25
35 moves	6	610	1,837	35

The reference count for the 21-move construction includes occurrences inside its 898 inline combinations. It therefore has 1,620 combination nodes altogether. The other two encodings have one combination per nonbase row. Their row numbers and orientation codes belong to three distinct tables.

6.1 Exact reconstruction

All calculations use finite sets of integer pairs. Each primary program parses its literal data, checks the grammar and backward references, applies the specified placements, and reconstructs every required set, envelope, and height. A separate implementation for each dialect independently parses and reconstructs the same data. These programs are supplied with the complete editable source; no strategy search, network connection, or undistributed game tree is needed. Appendix G gives the commands.

For the 21-move certificate the reconstruction includes every inline node, its enclosing coordinate frame, and the 32 children of the last card. It checks the endpoint, including containment of its 251-cell envelope in $\{0, \dots, 16\}^2$, the elementary cards, the successive intersections in Table 1, and the nonlocal continuation. There are 37,042 local reply classes over all combination nodes. For the 25-move certificate it checks all 2,010 reduced compositions, their normalized full cards, the required-size histogram, the five exceptional opening classes, and the elementary and nonlocal examples. For the 35-move certificate it checks all 616 templates, 26,703 local reply classes, the four terminal interfaces, and the longer tactical identities in Appendix B.

A local reply check uses the minimal Maker ownership required after the pivot; additional Maker cells only remove possible legal replies. One exterior case represents cells outside the parent envelope, since every child envelope lies inside it. In the reduced encoding this statement is applied after normalization. These finite checks support the set calculations. Their coverage of arbitrary ownership and the infinite board follows from the proved composition rule. Likewise, tests of sample opening coordinates supplement the symbolic whole-grid opening proof in Appendix B.

The supplied test harnesses compare the full reconstructed cards, not only terminal cardinalities. They also test rejection of malformed or truncated inputs and execution with Python optimization. Programs either use explicit exceptions for required checks or refuse execution when assertions have been disabled. The harnesses state their finite checks separately from the game-theoretic claims.

6.2 Literal identities and formal coverage

The literal data are printed in Appendices D–F. Their literal SHA-256 identities are

```
21: 3fa12d36a6d4dbb185e3f2808c8dfde13d85ee309afbca030d6ad17ae9fe3d04
25: 41cd04620af889771862dceda7383afd99933ae03ddb0a566a00d742a8249b0
35: 16d077f0b64df70dd67fb6e9651636d42eaab0ce0f01f46180a57846e38fedc2
```

The middle file includes its initial newline and all 41 index markers. The source guide also records the normalized numeric serialization of that table; it is a separate identity from the full literal. The programs read these files as data and do not execute them as code. Matching bytes do not replace reconstruction, and matching sets do not replace the semantic induction or the move-count argument.

Appendix C records the scope of the finite reconstruction supplement for the 35-move certificate. This supplement proves seven finite reconstruction claims. It does not prove a complete winning strategy, the 21- or 25-move construction, or the finite-board corollary. The finite Python reconstructions do not themselves establish a Lean theorem or kernel compilation. All three conventional proofs are given independently of this supplement.

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A The reduced-envelope 25-move construction

This construction records only those cells whose freedom from Breaker is not already implied by Maker’s required holdings. It gives a second finite strategy, with a different placement convention and a different opening cover. We first justify this representation and its exact relationship to Section 2, then reconstruct the certificate and explain its elementary and nonlocal attacks.

Throughout this appendix, a numbered 25-*card* belongs to the family

$$\mathcal{C}_i^{25} = (P_i, E_i, h_i), \quad 0 \leq i \leq 2015.$$

The symbols P_i, E_i, h_i in this appendix refer only to that family. Its card numbers, placement codes, and local coordinates are independent of the 21- and 35-move constructions. In particular, their cards with the same number need not have the same sets or meaning.

A.1 Reduced requirements and their normalization

Let $P, E \subset \mathbb{Z}^2$ be finite and let $h \geq 1$ be an integer. A *valid reduced card* (P, E, h) means the following: at every Maker turn with finite disjoint ownership sets M, B , if

$$P \subseteq M, \quad B \cap E = \emptyset, \quad (12)$$

Maker has already won or can win within h further actual Maker claims. We do not impose $P \subseteq E$, nor do we require P and E to be disjoint. Additional Maker cells and arbitrary Breaker cells outside E are allowed. A signed permutation and translation applied to both sets preserves validity and height, by the same transport argument as for full cards.

The normalization

$$(P, E, h) \mapsto (A, T, h) = (P, P \cup E, h) \quad (13)$$

preserves exactly the admissible positions. Indeed, $P \subseteq M$ and $M \cap B = \emptyset$ already imply $B \cap P = \emptyset$; under that assumption,

$$B \cap E = \emptyset \iff B \cap (P \cup E) = \emptyset.$$

Conversely, a full card (A, T, h) with $A \subseteq T$ has the reduced representative $(A, T \setminus A, h)$. Normalizing this representative returns (A, T, h) exactly. Normalizing an arbitrary (P, E, h) and then reducing replaces E by $E \setminus P$; this changes no hypothesis on a legal position. Thus disjoint reduced sets are convenient representatives, not an extra assumption in the semantics.

Proposition 8 (Reduced composition). *Let (P_j, E_j, h_j) , $1 \leq j \leq m$, be valid reduced cards already placed in one coordinate plane, where $1 \leq m < \infty$. Given an attack cell x , put*

$$\begin{aligned} U &= \bigcup_j P_j, & V &= \bigcup_j E_j, & I &= \bigcap_j E_j, \\ H &= U \cup I, & P &= H \setminus \{x\}, & E &= (V \setminus H) \cup \{x\}, & h &= 1 + \max_j h_j. \end{aligned} \quad (14)$$

Then (P, E, h) is valid. Its normalization is exactly the full card obtained by Lemma 3 from the normalized children $(P_j, P_j \cup E_j, h_j)$ at x ; in particular, normalization changes neither the required set nor the height.

Proof. We give the direct game argument, including the move that is needed when the attack cell is already owned. Suppose (12) holds for the parent and no target is complete. Because $x \in E$, Breaker does not own x . If it is free, claim it. Otherwise Maker already owns it and claims any free cell instead; finite ownership on $\mathbb{Z}^2$ guarantees that one exists. This replacement consumes one actual Maker claim, with its ensuing Breaker reply if play continues. In either case the new Maker set M' contains H , since it contains P and x .

Every older Breaker cell avoids every child envelope E_j . To see this without assuming $P_j \subseteq E_j$, note that the cells of E_j outside H lie in E , while its cells inside H now belong to Maker. If the move has not won, the next legal Breaker cell b is outside M' , hence outside $I \subseteq H$. Since the child list is nonempty, some E_j omits b . At the next Maker turn this child satisfies

$$P_j \subseteq U \subseteq M', \quad (B \cup \{b\}) \cap E_j = \emptyset.$$

It wins in at most h_j additional Maker claims. The total is at most $1 + h_j \leq h$. This also covers replies anywhere outside the finite sets displayed in the construction.

For the asserted exact agreement, the pointwise identity is

$$U \cup \bigcap_j (P_j \cup E_j) = U \cup \bigcap_j E_j = H. \quad (15)$$

For a point outside U , membership in $P_j \cup E_j$ is equivalent to membership in E_j for every j ; points inside U belong to both sides. This proves the identity without any containment or disjointness assumption on the child sets. Consequently the full composition's required set is $H \setminus \{x\} = P$. Its envelope is

$$\begin{aligned} \{x\} \cup \bigcup_j (P_j \cup E_j) &= \{x\} \cup U \cup V \\ &= (H \setminus \{x\}) \cup ((V \setminus H) \cup \{x\}) = P \cup E. \end{aligned}$$

Here $I \subseteq V$, using the nonempty list. Both height recursions are $1 + \max_j h_j$. $\square$

The same argument has a Breaker-turn form. If immediately before a Breaker reply Maker owns $P \cup \{x\}$ and Breaker avoids E , then after that reply a child applies within at most $\max_j h_j$ further Maker claims. The attack cell need not have been the latest Maker claim. This follows directly from ownership of H and the older Breaker avoidance proved above.

A.2 Bases, anchors, and the decimal certificate

All attacks in the local coordinates of a 25-card occur at $o = (0, 0)$. Order the target cells as

$$(L_0, \dots, L_5) = ((0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (4, 1)).$$

Thus $S = \{L_0, \dots, L_5\}$. The six base cards are

$$P_i = (S \setminus \{L_i\}) - L_i, \quad E_i = \{o\}, \quad h_i = 1, \quad 0 \leq i < 6. \quad (16)$$

If o is already Maker's, the translated target $S - L_i$ is already complete. Otherwise o is free and completes it in one move. Their normalized envelopes are exactly $S - L_i$.

For a reference to card j , order its required set P_j lexicographically, by the first coordinate and then the second, starting with index zero. Write a placement code r uniquely as $r = 8k + d$, $0 \leq d < 8$, and let $a_{j,k}$ be the k th cell in that order. The code is defined only when $0 \leq k < |P_j|$. Set

$$\Phi_{j,r}(v) = R_d(v - a_{j,k}), \quad (17)$$

where the signed permutations are given explicitly by

d	0	1	2	3
$R_d(x, y)$	(x, y)	(y, x)	$(-x, y)$	$(-y, x)$
d	4	5	6	7
$R_d(x, y)$	$(x, -y)$	$(y, -x)$	$(-x, -y)$	$(-y, -x)$

Equivalently, swap first when d is odd, then negate the first coordinate for the bit of value 2, and the second for the bit of value 4. In particular codes 3 and 5 use the swap-first order; they must not be interpreted using the 21-card decoder. Each map in (17) aligns one *required* child cell with the parent's attack o . The anchor is taken from P_j , not E_j or the normalized envelope. A child placement inside a current placement F is $F \circ \Phi_{j,r}$.

The complete literal certificate, printed in Appendix E and supplied as `verification/supporting/route25/certificate.txt`, has 2010 numeric lines defining cards 6 through 2015 in order. Blank lines have no effect. A marker `# n` asserts that the next card number is n and creates no card.

There are 41 such markers. Every numeric line consists of decimal blocks of exactly three digits. At current card i , read these blocks from left to right; a block with value b decodes as follows:

condition	referenced card j	placement code r
$0 \leq b < 240$	$\lfloor b/40 \rfloor$	$b \bmod 40$
$240 \leq b < 740$	$i - 1 - \lfloor (b - 240)/100 \rfloor$	$(b - 240) \bmod 100$
$740 \leq b \leq 999$	$10(b - 740) + \lfloor c/100 \rfloor$	$c \bmod 100$

(18)

The last case consumes the next three-digit block as c as well. A missing second block is invalid. Leading zeros are part of the three-digit format. Every resulting pair (j, r) must satisfy $0 \leq j < i$ and the anchor range in (17), and each numeric line must contain at least one pair.

For the decoded list $\mathcal{L}_i$ of card i , form the placed children

$$(\Phi_{j,r}(P_j), \Phi_{j,r}(E_j), h_j), \quad (j, r) \in \mathcal{L}_i,$$

and apply (14) at $x = o$ to define P_i, E_i, h_i . Every placed required set contains o , because of its anchor. The resulting P_i and E_i are disjoint, and $o \in E_i$. These properties are consequences of the recurrence; the soundness proof did not assume them. All references are backward, so induction from (16) proves the validity of every card whose line decodes. No game-tree search or additional compatibility test is needed: the union and intersection in the composition formula put all necessary extra ownership into the parent's required set.

For example, 193233 decodes into $(4, 33), (5, 33)$. The next line, 165272, decodes at $i = 7$ into $(4, 5), (6, 32)$. At $i = 2015$, 927400 is an extended term $(1874, 0)$, whereas 240 is the relative term $(2014, 0)$. Thus all three cases in (18) refer to the same anchor placement rule after decoding.

A.3 The first two composite cards

The first two rows exhibit both simultaneous finishes and transfer after a forced block. For card 6, the two codes $33 = 8 \cdot 4 + 1$ use anchors $(1, 0) \in P_4$ and $(-1, 0) \in P_5$, respectively. Their placed envelopes are the singletons $\{(0, -1)\}$ and $\{(0, 1)\}$. Their placed required sets are

$$\{(-1, t) : -4 \leq t \leq -1\} \cup \{o\}, \quad \{(-1, t) : -3 \leq t \leq 0\} \cup \{o\}.$$

The intersection of the two singleton envelopes is empty. Composition therefore gives exactly

$$P_6 = \{(-1, t) : -4 \leq t \leq 0\}, \quad E_6 = \{(0, -1), o, (0, 1)\}, \quad h_6 = 2. \quad (19)$$

After the attack at o , the two one-hole copies have different holes. If neither is already complete, a single Breaker reply cannot block both. All five cells of the required column are outside E_6 ; this is an explicit reason the reduced notation must allow $P_6 \not\subseteq E_6$.

For card 7, code 5 at card 4 uses the first lexicographic anchor $(-3, -1)$ and $R_5(x, y) = (y, -x)$. Its placed required set and hole are

$$\{(0, -3), (0, -2), (0, -1), o, (1, -4)\}, \quad \{(1, -3)\}.$$

Code $32 = 8 \cdot 4$ at card 6 uses anchor $(-1, 0)$ and is translation by $(1, 0)$. Its placed required set is $\{(0, t) : -4 \leq t \leq 0\}$, its envelope is $\{(1, -1), (1, 0), (1, 1)\}$, and its attack is $(1, 0)$. Again the two child envelopes are disjoint, giving

$$\begin{aligned} P_7 &= \{(0, -4), (0, -3), (0, -2), (0, -1), (1, -4)\}, \\ E_7 &= \{o, (1, -3), (1, -1), (1, 0), (1, 1)\}, \quad h_7 = 3. \end{aligned} \quad (20)$$

After the attack at o , failure to block $(1, -3)$ lets Maker complete the base child on the next move. A block there leaves the translated card 6 available, whose next attack is $(1, 0)$. If one of these attacks was already owned, the general card semantics either detects an earlier target or charges the required fresh replacement. The ensuing reply is never skipped. Figure 4 displays the two configurations.

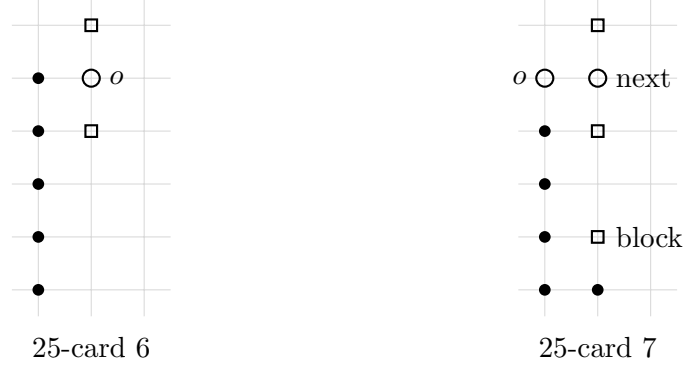


Figure 4: The elementary reduced cards. Solid disks are the required Maker cells. In card 6, attacking the open disk o gives the two square-marked finishes. In card 7, the attack at o threatens the square $(1, -3)$; after that block, the open disk $(1, 0)$ starts the translated card 6, with finishes at the other two squares.

A.4 The complete finite reconstruction and its endpoint

We now apply the same recurrence to the entire literal, retaining every row. The supplied primary program `verification/supporting/route25/verify_certificate.py` and independent program `verification/supporting/route25/independent_certificate.py` reconstruct its finite sets and heights. Their checks include the decimal grammar, every backward reference and anchor, the marker numbers, the required number of rows, and the exact endpoint. Malformed or truncated input is rejected; Appendix G gives the verification commands and their scope. Their finite arithmetic supports the set identities below; the reason these sets give a game strategy is Proposition 8 and the induction just proved, independently of either program.

The complete reconstruction has 2016 cards and 5862 placed child terms. The number of cards of each required-set size is

$ P_i $	0	1	2	3	4	5	6	7	8	9	10
number	1	5	52	217	448	523	416	219	103	30	2.

(21)

The last card can also be inspected through the much smaller terminal table. Its 28 children comprise exactly the following five groups in the printed order. Each listed code has anchor index $k = 0$. The last column gives the size of the intersection of all placed reduced envelopes through the end of that row of the table. The required sets of these five child cards are

$$P_{1874} = P_{1985} = P_{1996} = \{(-1, 0)\}, \quad P_{1998} = \{(0, 1)\}, \quad P_{2014} = \{(1, 0)\}. \quad (22)$$

Every placed requirement is therefore $\{o\}$. The table gives $U = \{o\}$ and $I = \emptyset$, hence $H = \{o\}$ and

$$P_{2015} = \emptyset, \quad |E_{2015}| = 389, \quad h_{2015} = 25. \quad (23)$$

Child j	Codes r	$ E_j $	h_j	Cumulative intersection
1874	0, 1, 2, 5	292	24	184
1985	0, 2, 4, 6	162	19	96
1996	0, 1, 2, 5	117	15	49
1998	0, 1, 2, 3, 4, 5, 6, 7	119	20	32
2014	0, 1, 2, 3, 4, 5, 6, 7	76	14	0

Table 2: All children of 25-card 2015. The common reduced envelope disappears after the last group.

The last height is $1 + 24$, not a bound obtained by identifying this certificate with either of the other two constructions.

For an explicit description of the final finite region, its section at ordinate y is empty when $|y| > 11$; otherwise it is the set of integer abscissae in the following table:

$ y $	Abscissae x	$ y $	Abscissae x
0	$[-10, 10]$	6	$[-9, 9]$
1	$[-11, 11]$	7	$[-7, 7]$
2	$[-10, 10]$	8	$[-6, 6]$
3	$[-11, 11]$	9	$[-6, 6]$
4	$[-10, 10]$	10	$[-5, 5]$
5	$[-10, 10]$	11	$\{-3, -1, 1, 3\}$

Here intervals mean all integer points. The row counts give $21 + 2(23 + 21 + 23 + 21 + 21 + 19 + 15 + 13 + 13 + 11 + 4) = 389$.

Proposition 9 (The 25-move strategy). *There is one ordinary Maker policy on the infinite square board that, from the empty position, completes a copy of S within 25 actual fresh Maker claims against every legal Breaker continuation. In the fixed-endpoint formulation, after extending the policy by fresh moves beyond an earlier win, the Maker set after its 25th claim contains a target whenever the first 24 Breaker replies are legal along the policy.*

Proof. Induction using (16) and Proposition 8 proves the validity of all 2016 cards. Equation (23) therefore applies at the empty position. To fix one policy before the opponent's choices, fix an enumeration of $\mathbb{Z}^2$ and the printed order of every child list. Start with card 2015 in its identity placement. At a composite card in placement F , take $F(o)$ if free, or the first free cell of the enumeration if $F(o)$ is already owned. Stop at a target. If play continues, after the next legal Breaker reply choose the first child (j, r) whose placed reduced envelope avoids the current Breaker set. The composition proof guarantees its existence and its required ownership. Replace the active placement by $F \circ \Phi_{j,r}$. At a base, its hole is either already filled, giving a target, or can be taken to finish one.

Each nonterminal descent consumes one actual fresh Maker claim, including every replacement, and decreases the height. No path from height 25 needs more than 25 such claims. As in Section 4, the active card and placement are determined by replaying the complete history. On a history inconsistent with the policy, or beyond a completed target, prescribe the first free cell instead. This makes the policy globally legal on finite legal histories and fixes it independently of future replies.

If the first win is on claim $m \leq 25$, exactly $m - 1$ replies have preceded it. Thus only 24 replies can be needed. Under the extended policy an earlier target persists, proving the fixed-endpoint statement without any assumption on a 25th Breaker reply. $\square$

A.5 Why only five first-reply classes need extra cards

The final row can be understood geometrically as a cover of all first Breaker replies. After Maker takes o , each of the eight placements $\Phi_{2014,d}$, $0 \leq d < 8$, has required set $\{o\}$. Let D be the intersection of their reduced envelopes. The exact finite calculation is

$$D = \{(\pm t, 0), (0, \pm t) : t = 1, 2, 3\} \cup \{(\pm 1, \pm 1)\} \cup \{(\pm 2, \pm 1), (\pm 1, \pm 2)\}, \quad (24)$$

with independent signs. It contains 24 cells. In particular the sorted absolute coordinates, larger first, have exactly the five values

$$(1, 0), \quad (1, 1), \quad (2, 0), \quad (2, 1), \quad (3, 0).$$

The set D is invariant under all signed permutations: composing any one of those permutations with the eight placements merely permutes them. Equation (24) is a finite set identity, so it covers unbounded first replies as well: for every $b \notin D$, at least one of the eight envelopes omits b . That placement of card 2014 is already available, giving at most 14 further claims.

For the five remaining classes, the following placements omit the listed first Breaker cell. All use $k = 0$ and have placed required set $\{o\}$. For example, the first placement is $\Phi_{1874,2}(x, y) = (-x - 1, y)$

First Breaker cell	Child j	d	Next attack	Further bound h_j
(1, 0)	1874	2	(-1, 0)	24
(1, 1)	1985	6	(-1, 0)	19
(2, 0)	1996	2	(-1, 0)	15
(2, 1)	1998	2	(0, -1)	20
(3, 0)	1996	2	(-1, 0)	15

Table 3: The five exceptional opening classes for the 25-move construction. Each bound begins at Maker's second turn.

because its anchor is $(-1, 0)$; the fourth is $\Phi_{1998,2}(x, y) = (-x, y - 1)$ because its anchor is $(0, 1)$. These formulas give the displayed attacks directly. For an arbitrary reply in one of the five classes, choose a signed permutation Q taking its representative to that reply and use the placement $Q \circ \Phi_{j,d}$. It still requires only o , and its envelope omits that reply. Thus the table and the eight placements of card 2014 give a direct whole-board opening proof, with largest total $1 + 24 = 25$. The terminal row in Table 2 encodes a finite opening cover of this kind using its specified 28 placements.

A.6 Two continuations that leave the initial line

We finish by following two actual children of card 1874. This makes the composition of coordinate maps visible and shows why an implementation must retain the complete envelopes. Suppose Maker has played o , Breaker has replied $(1, 0)$, and Maker uses the first row of Table 3 to attack $(-1, 0)$. The current outer placement is

$$F(x, y) = \Phi_{1874,2}(x, y) = (-x - 1, y), \quad M = \{o, (-1, 0)\}, \quad B = \{(1, 0)\}$$

before Breaker's next reply. The two following continuations are present in the literal row for card 1874:

Next reply	Child j	Code r	Anchor $a_{j,k}$	Next attack
$(-1, -1)$	1716	8	$(-2, -1)$	$(-3, 1)$
$(-2, 0)$	1752	3	$(-2, 0)$	$(-1, 2)$

For the first, the native required set is $P_{1716} = \{(-3, -1), (-2, -1)\}$. Since $8 = 8 \cdot 1 + 0$, the lexicographic anchor is the second point, $(-2, -1)$, and

$$\Phi_{1716,8}(x, y) = (x + 2, y + 1), \quad (F \circ \Phi_{1716,8})(x, y) = (-x - 3, y + 1).$$

The transformed requirement is precisely $\{o, (-1, 0)\}$. The reduced envelope has 256 cells and satisfies the exact check

$$(F \circ \Phi_{1716,8})(E_{1716}) \cap \{(1, 0), (-1, -1)\} = \emptyset. \quad (25)$$

Thus the old reply as well as the new reply is excluded, and the transformed pivot is $(-3, 1)$. It is distinct from all four cells then owned by either player. Card 1716 supplies a height-23 continuation after the first two Maker claims, consistent with the total bound 25.

For the second, $P_{1752} = \{(-2, 0), (-2, 1)\}$ and $r = 3$ selects the first anchor and the swap-first map $R_3(x, y) = (-y, x)$. Hence

$$\Phi_{1752,3}(x, y) = (-y, x + 2), \quad (F \circ \Phi_{1752,3})(x, y) = (y - 1, x + 2).$$

Its requirement is again exactly $\{o, (-1, 0)\}$, while its 93-cell reduced envelope satisfies

$$(F \circ \Phi_{1752,3})(E_{1752}) \cap \{(1, 0), (-2, 0)\} = \emptyset. \quad (26)$$

Its pivot $(-1, 2)$ is free, and this child has height 15. Equations (25)–(26) use the complete reconstructed envelopes, not just the new pivot or the required pair of Maker cells.

The two replies here are legal examples, not forced Breaker choices. For either example the indicated child is usable; the globally fixed first-surviving-child policy may select an earlier usable child in the same row. Other replies are covered by the full child list and Proposition 8. Neither the game nor this certificate restricts Maker's holdings to a connected shape: these two attacks deliberately leave the initial horizontal pair.

B The 35-move construction and its conditional interfaces

This appendix gives a separate construction from six immediate-completion cards and 610 further rows. Besides an ordinary 35-move strategy, it provides four conditional strategies with small required sets and specified regions in which Breaker may already have played. Those conditional conclusions concern arbitrary prior ownership, so they do not follow merely from a better bound for the empty board. We retain both the complete opening and the transfer, opposite-tip, and bent-shape continuations that explain how this certificate operates.

Throughout this appendix, $A_i^{35}, T_i^{35}, h_i^{35}$ denote the required set, envelope, and height of row i of the *35-move certificate*. These indices and the placement codes below are separate from those of the 21- and 25-move constructions. The meaning of a card is the arbitrary-ownership meaning of Section 2: at any Maker turn, finite disjoint sets M, B with $A_i^{35} \subseteq M$ and $B \cap T_i^{35} = \emptyset$ permit a win within h_i^{35} further actual Maker claims, unless a target is already complete. Additional ownership is allowed everywhere.

B.1 The letter-and-offset certificate

Write the target points in the fixed order

$$s_0 = (0, 0), \quad s_1 = (1, 0), \quad s_2 = (2, 0), \quad s_3 = (3, 0), \quad s_4 = (3, 1), \quad s_5 = (4, 1), \quad o = (0, 0).$$

The six initial cards are

$$T_i^{35} = S - s_i, \quad A_i^{35} = T_i^{35} \setminus \{o\}, \quad h_i^{35} = 1 \quad (0 \leq i \leq 5). \quad (27)$$

They are valid by immediate completion. If the missing point is already Maker-owned, the target has already been obtained.

For $i = 6, \dots, 615$, the literal row consists of i followed by a nonempty list of placement terms. A term $\mathbf{kLuv}$ contains an earlier decimal row index $k < i$, one letter L , and two individual decimal digits u, v . Its placement is

$$F_{Luv}(x, y) = R_L(x, y) + (u - 4, v - 4), \quad (28)$$

with the eight maps specified by

L	a	b	c	d
$R_L(x, y)$	(x, y)	(y, x)	$(x, -y)$	$(-y, x)$
L	e	f	g	h
$R_L(x, y)$	$(-x, y)$	$(y, -x)$	$(-x, -y)$	$(-y, -x)$

Thus **614b43** means $(x, y) \mapsto (y, x - 1)$, including its translation; it is not one of the decimal orientation codes used in the other two certificates.

If the terms of row i are $(k_j, L_j u_j v_j)$, put $C_j = F_{L_j u_j v_j}(A_{k_j}^{35})$ and $D_j = F_{L_j u_j v_j}(T_{k_j}^{35})$. Define

$$\begin{aligned} T_i^{35} &= \{o\} \cup \bigcup_j D_j, \\ A_i^{35} &= \left(\bigcup_j C_j \cup \bigcap_j D_j \right) \setminus \{o\}, \\ h_i^{35} &= 1 + \max_j h_{k_j}^{35}. \end{aligned} \quad (29)$$

Lemma 3 and induction on i prove every card defined in this way. The inductive argument is valid for every syntactically correct list of backward references; the particular list determines the useful required sets and envelopes.

For clarity, the complete finite substitution can be performed as follows. Initialize the six triples by (27). Read each remaining row in increasing order, reject an empty list or a reference outside $0, \dots, i - 1$, decode each affine map by (28), and apply it to both stored sets. Take the union and intersection in (29), store the resulting triple, and proceed to the next row. All coordinates are integers and all sets are finite. At the end of each iteration the stored triples are exactly the triples defined by the displayed equations, which is the loop invariant justifying the reconstruction.

The complete 610-row literal is printed in Appendix F and supplied as `verification/supporting/prior-35/certificate.txt`; it has one row per line, including the final newline. The primary and two independent reconstructions are described in Appendix G. They use, respectively, integer matrices and ordinary set operations, quarter turns and incidence masks, and signed coordinate lookups with cellwise membership tests. Thus no search procedure or undisclosed strategy tree is needed to recover the certificate.

B.2 The first two compositions: five cells in a row

The first two nonbase rows are

$$6 \text{ 0a03}, \quad 7 \text{ 4a34 6a54}.$$

Set

$$L_- = \{-4, -3, -2, -1\} \times \{-1\}, \quad L_+ = \{-3, -2, -1, 0\} \times \{-1\}.$$

Row 6 has

$$T_6^{35} = S + (-4, -1) = L_- \cup \{(-1, 0), o\}, \quad A_6^{35} = L_- \cup \{(-1, 0)\}, \quad h_6^{35} = 2.$$

The height is the recursively assigned upper bound: after claiming its pivot, Maker has in fact completed this target immediately.

The two placed children in row 7 have

term	required set	envelope
4a34	$L_- \cup \{o\}$	$L_- \cup \{(-1, 0), o\}$
6a54	$L_+ \cup \{o\}$	$L_+ \cup \{o, (1, 0)\}.$

(30)

Their common envelope is $(\{-3, -2, -1\} \times \{-1\}) \cup \{o\}$, already contained in the union of their required sets. Consequently

$$A_7^{35} = \{-4, -3, -2, -1, 0\} \times \{-1\}, \quad T_7^{35} = A_7^{35} \cup \{(-1, 0), o, (1, 0)\}, \quad h_7^{35} = 3.$$

After Maker owns the five lower cells and the pivot o , either $(-1, 0)$ or $(1, 0)$ completes one of the two displayed copies. Both lie in the parent envelope, so neither is occupied by an earlier Breaker stone. If neither target is already complete, Breaker can claim at most one of these two cells. This is the elementary double threat that the longer compositions reuse. Its direct two-claim bound is sharper than the stored height 3; the certificate deliberately uses the uniform recursion (29).

B.3 Four conditional outputs

The following finite identities supply the opening. A set in the last column is wholly disjoint from the envelope; it is not asserted to be the entire complement of that envelope.

Proposition 10. *The 35-move certificate has 616 cards, including its six base cards, and 1,837 placement terms. Every card satisfies $A_i^{35} \subsetneq T_i^{35}$, $o \in T_i^{35} \setminus A_i^{35}$, and $h_i^{35} \leq 34$. Four of its conditional interfaces are*

i	A_i^{35}	$ T_i^{35} $	h_i^{35}	region disjoint from T_i^{35}
557	$\{(-1, 0)\}$	118	19	$\{(x, y) : x \leq -3, y \leq -2\}$
595	$\{(-1, 0)\}$	222	27	$\{(x, y) : x = -2, y \leq -2\}$
603	$\{(-1, 0)\}$	275	27	$\{(x, y) : x \leq -3, y = 0\}$
615	$\{(1, 0)\}$	686	34	$\{(1, -1)\}.$

(31)

Each row applies at any Maker turn with its required point owned and its entire envelope free of Breaker, regardless of all other ownership. For row 615 there is also a Breaker-turn conclusion: if Maker owns $\{o, (1, 0)\}$ and Breaker avoids T_{615}^{35} , then after any legal next reply Maker wins within 33 further claims, unless already victorious.

Proof. The finite identities follow by the substitutions (27)–(29). The delivered reconstructions check every row index, every placement and backward reference, the full sets and heights, and the four outputs in (31). An omitted infinite region is checked by testing its defining predicate at *every* point of the finite envelope. For example, the first check verifies $x > -3$ or $y > -2$ for every $(x, y) \in T_{557}^{35}$; this proves disjointness from the whole infinite region, without sampling it.

For each composed card the reconstruction also verifies $C_j \subseteq A_i^{35} \cup \{o\}$ and $D_j \subseteq T_i^{35}$. It checks each possibly legal local reply in $T_i^{35} \setminus (A_i^{35} \cup \{o\})$, and one extra class representing all cells outside T_i^{35} . There are 26,703 such classes in total over the 610 composed rows. In each class some child envelope is missed. The exterior class is exact because every child envelope is contained in the parent envelope. Extra Maker ownership can only remove legal reply cells, and earlier Breaker ownership is excluded by the envelope hypothesis. These checks establish the stated finite identities; the game-theoretic conclusion is the induction using Lemma 3, not an inference from the number of tests. Its Breaker-turn assertion gives the final claim, since $h_{615}^{35} - 1 = 33$. $\square$

B.4 The complete ordinary opening

Theorem 11. *There is one ordinary Maker policy on the initially empty infinite square board that completes an allowed copy of S within 35 actual Maker claims against every legal Breaker play. Equivalently, after extending a won position by fresh moves if necessary, its 35th Maker set contains a target whenever the first 34 Breaker replies are legal. No 35th Breaker reply is required.*

Proof. Maker first claims the actual origin. Write $q_* \neq o$ for Breaker's first cell, and let $0 \leq a \leq b$ be the sorted absolute values of its two coordinates. Since the cell differs from the origin, $b \geq 1$. We first handle all unbounded distant replies, and then the eight adjacent or diagonal replies.

Suppose $b \geq 2$. In local coordinates put $p = (-1, 0)$, and choose

$$(i, d) = \begin{cases} (557, (-a, -b)), & a \geq 2, \\ (595, (-1, -b)), & a = 1, \\ (603, (-b, 0)), & a = 0. \end{cases} \quad (32)$$

In each case d and q_* have the same sorted absolute coordinates, so there is a signed permutation $R \in \mathcal{G}$ with $Rd = q_*$. Use the affine map

$$F(z) = R(z - p). \quad (33)$$

It sends the required cell p to Maker's first cell, and the local Breaker cell $p + d$ to q_* . In the three respective cases,

$$p + d = (-a - 1, -b), \quad (-2, -b), \quad (-b - 1, 0).$$

The first has $x \leq -3, y \leq -2$, the second has $x = -2, y \leq -2$, and the third has $x \leq -3, y = 0$. Thus it lies in the corresponding omitted region in (31). At Maker's second turn, $F(A_i^{35})$ is owned and $F(T_i^{35})$ contains no Breaker cell. The placed conditional card applies, giving totals at most $1 + 19 = 20$, $1 + 27 = 28$, and $1 + 27 = 28$, respectively. The argument used no upper bound on a or b .

It remains to treat $b = 1$. Put

$$q = (1, -1), \quad P = \{o, (1, 0)\}, \quad p = \begin{cases} (1, 0), & a = 0, \\ o, & a = 1. \end{cases}$$

For the coordinate-adjacent reply, $q - p = (0, -1)$; for the diagonal reply, $q - p = (1, -1)$. In either case choose $R \in \mathcal{G}$ with $R(q - p) = q_*$, and again use (33). Then $F(p) = o$ and $F(q) = q_*$.

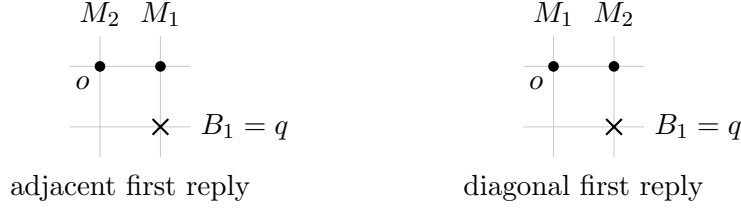


Figure 5: Local coordinates immediately after Maker's second claim and before Breaker's second reply. Dots are Maker's domino P , the cross is the existing Breaker cell q , and subscripts show the order of the actual moves. Row 615 controls the next reply in both cases.

Maker's second move claims the other cell of $F(P)$. It is fresh: the two cells of P differ, and $q \notin P$. Immediately after this move,

$$M = F(P) = F(A_{615}^{35} \cup \{o\}), \quad B = \{F(q)\}, \quad B \cap F(T_{615}^{35}) = \emptyset.$$

The two Maker cells have been obtained with the actual first Breaker reply between them. In the adjacent case the local pivot o was played second; in the diagonal case it was played first. This order does not matter to the state-based Breaker-turn assertion of Proposition 10.

It is now Breaker's second turn. Every legal reply misses at least one child envelope of the placed row 615, with that child's required set already owned. The child wins within at most 33 further Maker claims. Thus the total in the near case is

$$2 + 33 = 35.$$

Figure 5 records both possible orders of the domino.

Fix the choices before play: order the eight signed permutations and the terms in every row, and fix an enumeration of $\mathbb{Z}^2$. Use the first admissible opening matrix and the first child envelope missed by the latest reply. Compose each child map with the current affine frame. If its pivot is already Maker-owned and no target is complete, claim the first genuinely free cell in the enumeration; that replacement and the following legal reply are charged in the height recursion. Every child has smaller index, so the recursion terminates within the stated height. On histories inconsistent with earlier prescriptions, use the first free cell. This gives one total fresh policy, with the same early-stopping and extension conventions as Section 4. An extended 35-claim play contains only 34 intervening Breaker replies, proving the precise endpoint. $\square$

B.5 Transferring a four-cell attack to a new pair

We now examine the longer tactics inside the certificate. They explain how a small required set can support a continuation even after several forced parries. The conditions always concern the entire envelope; the visible stones alone do not license a card.

Put $V = \{-1\} \times \{-4, -3, -2, -1\}$. The relevant literal rows and reconstructed outputs are

$$\begin{array}{c}
 588 \ 3b33 \ 586a45, \quad 589 \ 4b43 \ 588a45, \\
 \begin{array}{c|c|c|c}
 i & A_i^{35} & |T_i^{35}| & h_i^{35} \\
 \hline
 586 & \{(0, -2), (0, -1)\} & 198 & 26 \\
 588 & (V \setminus \{(-1, -1)\}) \cup \{(0, -1)\} & 202 & 27 \\
 589 & V & 204 & 28.
 \end{array}
 \end{array} \tag{34}$$

x	$\{y : (x, y) \in T_{586}^{35}\}$	x	$\{y : (x, y) \in T_{586}^{35}\}$	x	$\{y : (x, y) \in T_{586}^{35}\}$
-7	$[3, 4]_{\mathbb{Z}} \cup \{6\}$	0	$[-2, 9]_{\mathbb{Z}}$	7	$[-5, 5]_{\mathbb{Z}}$
-6	$\{1\} \cup [3, 6]_{\mathbb{Z}}$	1	$[-6, 9]_{\mathbb{Z}}$	8	$[-2, 4]_{\mathbb{Z}}$
-5	$[-1, 9]_{\mathbb{Z}}$	2	$[-7, 9]_{\mathbb{Z}}$	9	$[-2, 4]_{\mathbb{Z}}$
-4	$[-1, 8]_{\mathbb{Z}}$	3	$[-6, 9]_{\mathbb{Z}}$	10	$[-2, 0]_{\mathbb{Z}}$
-3	$[-1, 11]_{\mathbb{Z}}$	4	$[-7, 7]_{\mathbb{Z}}$	11	$\{-2, 0\}$
-2	$[-2, 10]_{\mathbb{Z}}$	5	$[-5, 6]_{\mathbb{Z}}$		
-1	$[-1, 11]_{\mathbb{Z}}$	6	$[-5, 6]_{\mathbb{Z}}$		

Table 4: The complete 198-cell envelope of row 586. The translated envelopes governing the two forced parries and the transferred pair follow from (35).

Their envelopes satisfy the more informative identities

$$\begin{aligned} T_{588}^{35} &= (T_{586}^{35} + (0, 1)) \cup V, \\ T_{589}^{35} &= (T_{586}^{35} + (0, 2)) \cup (\{-1\} \times \{-4, -3, -2, -1, 0\}) \cup \{(0, -1)\}. \end{aligned} \quad (35)$$

In particular, each smaller continuation envelope is contained in the envelope under which the tactic starts.

For an explicit description of these envelopes, Table 4 gives every vertical fiber of T_{586}^{35} . In this table $[a, b]_{\mathbb{Z}}$ means every integer from a to b , and unlisted fibers are empty. Together with (35), it specifies all three complete envelopes, including cells far from the initial column.

Suppose $V \subseteq M$ and $B \cap T_{589}^{35} = \emptyset$ at a Maker turn. Claim o (or make the charged fresh replacement if it is already owned). The set

$$Q_1 = V \cup \{o, (0, -1)\}$$

is the target in the placed base card **4b43**. Its only possibly missing Maker cell is $(0, -1)$. If this point is already owned, Maker has won. Otherwise it is free, because it lies in the parent envelope. A Breaker reply other than $(0, -1)$ leaves that immediate completion available, so any defense avoiding this win must claim $(0, -1)$.

The other child **588a45** uses the map $z \mapsto z + (0, 1)$. Its required set is

$$(\{-1\} \times \{-3, -2, -1\}) \cup \{o\},$$

which is contained in the Maker set. Its envelope $T_{588}^{35} + (0, 1)$ is a subset of T_{589}^{35} and omits $(0, -1)$. All Breaker cells, including that first parry, therefore avoid it. Claim its pivot $(0, 1)$, with the same charged-replacement convention if necessary. Now

$$Q_2 = \{(-1, -3), (-1, -2), (-1, -1), (-1, 0), o, (0, 1)\}$$

is complete except possibly at $(-1, 0)$. It is another allowed target: $Q_1 = R_b(S) + (-1, -4)$ and $Q_2 = R_b(S) + (-1, -3)$. Thus, unless Maker has already won or can complete Q_2 on the next move, Breaker must parry at $(-1, 0)$.

After these two parries, the continuation **586a45** of the translated row 588 is row 586 translated by $(0, 2)$. Its required set is exactly $\{o, (0, 1)\}$, now owned. Its envelope

$$D = T_{586}^{35} + (0, 2)$$

lies in the original envelope and contains neither parry: $(0, -1) \notin D$ and $(-1, 0) \notin D$. These omissions follow directly from the fibers $x = 0$ and $x = -1$ in Table 4. The envelope of the preceding

row-588 continuation likewise omits the first parry by (35). Row 586 therefore applies at the next Maker turn, with pivot $(0, 2)$ and bound 26. The two earlier claims and their forced replies have transferred the attack from the original column to a new vertical pair, at total cost at most $1 + 1 + 26 = 28$ Maker claims.

This argument preserves every earlier Breaker restriction: all earlier Breaker cells lie outside the original envelope and hence outside every continuation envelope. Extra Maker cells may complete a target early or occupy one of the prescribed pivots; in the latter case a genuinely fresh replacement uses the same charged turn. No assumption that the displayed stones are the only owned stones is needed.

B.6 Opposite-tip continuations from four in a row

Row 612 consists exactly of

$$612 \quad 589b55 \quad 589h03.$$

Writing

$$W = \{-3, -2, -1, 0\} \times \{0\}, \quad G_+(x, y) = (y + 1, x + 1), \quad G_-(x, y) = (-y - 4, -x - 1),$$

the two placed requirements are both W , and their envelopes obey

$$G_+(T_{589}^{35}) \cap G_-(T_{589}^{35}) = W. \tag{36}$$

Consequently

$$A_{612}^{35} = W \setminus \{o\}, \quad T_{612}^{35} = G_+(T_{589}^{35}) \cup G_-(T_{589}^{35}), \quad |T_{612}^{35}| = 404, \quad h_{612}^{35} = 29.$$

After Maker adds the pivot, W is owned. A legal Breaker reply therefore cannot belong to both child envelopes; choose a child whose envelope it misses. All earlier Breaker cells avoid both envelopes by the parent hypothesis, and the full required column of the selected copy of row 589 is already owned. Its next pivot is either $(1, 1)$ or $(-4, -1)$, so the two attacks leave from opposite ends of the four-cell row. The transferred pair in the selected frame is $G_+(\{o, (0, 1)\})$ or $G_-(\{o, (0, 1)\})$, respectively. Identity (36) is what ensures a safe continuation against every reply, including replies far from the displayed four stones.

The overlap identity can also be read directly from Table 4 and (35). The vertical supports of the two placed envelopes are $[-6, 12]_{\mathbb{Z}}$ and $[-12, 6]_{\mathbb{Z}}$, respectively. On their common possible levels $-6 \leq y \leq 6$, the fibers have separated extrema for every $y \neq 0$: the rightmost point of the G_- fiber is strictly left of the leftmost point of the G_+ fiber. At $y = 0$, the G_+ fiber is $[-3, 14]_{\mathbb{Z}}$ and the G_- fiber is $[-17, 0]_{\mathbb{Z}}$. Their overlap is $[-3, 0]_{\mathbb{Z}}$. This gives exactly W , and also explains the envelope count $204 + 204 - 4 = 404$.

B.7 The exceptional bent-shape continuation

The near opening reaches the local Breaker-turn position with $P = \{o, (1, 0)\}$ owned and $q = (1, -1)$ occupied by Breaker. Here the twelve child terms of row 615 are

$$\begin{array}{cccc} 602h64 & 614b43 & 533f64 & 590h64 \\ 611g34 & 587d34 & 601f64 & 406b64 \\ 591d34 & 586d34 & 584f64 & 406d34. \end{array}$$

row-614 term	term after K	next pivot	envelope size	height
608b75	608a56	(1, 2)	354	32
543d56	543c64	(2, 0)	55	11
370g34	370h42	(0, -2)	35	11
392h34	392g42	(0, -2)	41	11
610d56	610c64	(2, 0)	343	32
600a64	600b45	(0, 1)	259	29
613a64	613b45	(0, 1)	563	30.

Table 5: All seven continuations after the exceptional pivot. Every placed child requires exactly $Q = \{o, (1, 0), (0, -1)\}$. Several children have the same pivot but different complete envelopes.

For each term let C_j, D_j be its placed requirement and envelope. Every C_j equals P , all D_j avoid q , and their full intersection is P . On deleting the second term, the intersection becomes

$$\bigcap_{j: \text{term}_j \neq 614b43} D_j = P \cup \{(-1, 0)\}. \quad (37)$$

The next pivots in these eleven children are end extensions $(-1, 0)$ or $(2, 0)$. Since Breaker cannot claim a point of P , one of them survives every next reply except $(-1, 0)$. Against that exceptional reply, the only surviving child is 614b43. Its map and pivot are

$$K(x, y) = (y, x - 1), \quad K(o) = (0, -1).$$

The pivot is fresh in the near-opening position: it differs from both Maker points and both Breaker points $q, (-1, 0)$.

More explicitly,

$$A_{614}^{35} = \{(1, 0), (1, 1)\}, \quad K(A_{614}^{35}) = P, \quad |T_{614}^{35}| = 678, \quad h_{614}^{35} = 33.$$

The complete envelope $K(T_{614}^{35})$ lies in T_{615}^{35} and avoids both existing Breaker cells. An exact expression for the difference is

$$\begin{aligned} T_{615}^{35} \setminus K(T_{614}^{35}) = \{ & (-15, 0), (-15, 2), (-14, 0), (-14, 1), \\ & (-14, 2), (-13, -4), (-13, 4), (-1, 0)\}. \end{aligned} \quad (38)$$

Thus the exceptional reply is excluded by the *full* continuation envelope, not merely by the three-cell configuration that Maker will obtain.

After Maker claims $(0, -1)$, write $Q = P \cup \{(0, -1)\}$. The seven children of row 614, composed with K , are listed in Table 5. In the table the term in the second column is expressed directly in the local coordinates of the near opening. For example, $K \circ F_{b75}(x, y) = (x + 1, y + 2) = F_{a56}(x, y)$.

Let C'_j, D'_j be the placed sets of these seven terms. Their reconstructed identities are

$$C'_j = Q \quad \text{for every } j, \quad \bigcap_{j=1}^7 D'_j = Q, \quad D'_j \subseteq K(T_{614}^{35}) \quad \text{for every } j. \quad (39)$$

The last inclusion excludes both old Breaker stones from every child envelope. Any legal third Breaker reply lies outside Q , so the intersection identity supplies a child envelope it misses. Its required set is already owned, and its height is at most 32. The selected pivot belongs to that envelope and is outside Q ; hence neither an old Breaker cell nor the new reply occupies it. It is

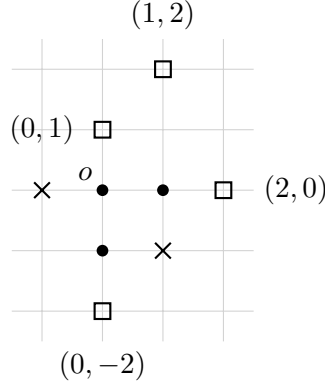


Figure 6: After the exceptional reply $(-1, 0)$ and Maker’s claim $(0, -1)$, the filled dots form Q and the crosses are the two Breaker cells. Hollow squares mark possible pivots after Breaker’s next reply. The diagram displays their geometry; their availability is determined by the seven envelopes specified by the terms in Table 5.

therefore fresh in this exact opening position. The full conditional version permits extra Maker ownership and uses a charged replacement if that pivot was owned previously.

The four distinct possible pivots are $(0, 1)$, $(0, -2)$, $(2, 0)$, $(1, 2)$, as shown in Figure 6. They are alternatives selected *after* the next Breaker reply, rather than a prescribed sequence of four Maker moves. The seven distinct envelopes in (39) determine which alternative is valid; the visible bent shape by itself does not justify any arbitrary choice. The bound after the exceptional second reply is $1 + 32 = 33$ further Maker claims, including the bent-shape pivot, which agrees with the $2 + 33 = 35$ opening calculation.

C Exact scope of the finite reconstruction supplement

This appendix records the scope of the finite reconstruction supplement for the 35-move certificate. The conventional proofs in this article are independent of this supplement.

C.1 The finite reconstruction theorem

The finite reconstruction supplement linked from `../..../lean/docs/187.md`, relative to the article root, has the endpoint `OAI.SnakyCertificate.certificate_correct`. Its definitions fix the target, six ordered base points, eight letter-order orientations, offset-by-four translation digits, and full-envelope recurrence of Appendix B. The formal envelope symbol H denotes T_i^{35} in that appendix. The theorem has seven clauses:

1. The table has exactly 610 nonbase rows, indexed 6 through 615, each with a nonempty list of allowed placements referencing earlier rows.
2. There are exactly 1,837 ordered placement terms, counted with multiplicity.
3. Row 557 has required set $\{(-1, 0)\}$, envelope size 118, height 19, and no envelope cell with $x \leq -3$ and $y \leq -2$.
4. Row 595 has required set $\{(-1, 0)\}$, envelope size 222, height 27, and no envelope cell with $x = -2$ and $y \leq -2$.

5. Row 603 has required set $\{(-1, 0)\}$, envelope size 275, height 27, and no envelope cell with $x \leq -3$ and $y = 0$.
6. Row 615 has required set $\{(1, 0)\}$, envelope size 686, height 34, and omits $(1, -1)$ from its envelope.
7. Every row 0 through 615 satisfies $A_i^{35} \subseteq T_i^{35}$, $(0, 0) \in T_i^{35} \setminus A_i^{35}$, and $h_i^{35} \leq 34$.

The last clause makes the required-set containment proper. The literal entries in the Lean source agree with the printed letter-and-offset certificate. This is a finite reconstruction statement, not a game-winning theorem: it does not include the opening, the arbitrary-ownership and Breaker-turn interfaces, policy extraction, or the additional tactical conclusions of Appendix B. The conventional arguments supply these game-theoretic implications.

The finite Python reconstructions described in this article are separate from Lean kernel checking; they do not themselves establish a Lean theorem or kernel compilation. The certificate copy under `verification/supporting/prior-35/` contains only the literal certificate data.

D Complete certificate data

The following listing contains every line of the certificate used in Proposition 5, without omissions. Cards $0, \dots, 5$ are the six bases (4); the listing starts at card 6. The grammar and bit-coded placements are specified in Section 3. A long logical line may wrap visually. The editable file `verification/certificate.txt` has one logical line per numbered card. Parentheses describe inline applications of exactly the same combination rule, with coordinates in the enclosing line's frame.

```

6 14 4:100 5:101
7 05 6:001 6:405
8 41 6:101 6:301
9 15 6:011 6:615
10 14 5:714 8:101
11 11 0:305 5:304
12 15 7:616 8:102 9:626
13 15 7:211 8:102 9:626
14 15 7:616 8:102 9:001
15 14 0:100 0:715
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 2:437 (74 54:273 158:324 (64 6:774 145:043))) (56 3:276 (43 54:657 247:327)))
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 156:435)))) (53 227:031 245:765 274:427) 559:405
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 (36 15:437 (33 11:637 38:746)))) 561 562:001

564 44 119:111 497:240 498 (54 12:113 12:315 79:121 135:112 182:316 (55 44:124 (24 5:024 453:253) (56 67:262 134:573 180:573 327:797 504:022 558:574 (57 16:254 235:668 (45 0:758 (25 102:564 (35 5:025 55:252) (24 5:254 342:252)))))) 560:111 563:103))

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590 23 52:010 588

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 398:777 (57 10:658 34:337) 628:033)
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 449:328 510:328 (54 117:123 134:337 180:337 337:669 341:689 435:327 558:336) (45 1:025 651:112) (45
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174:44E 475:057 475:297 483:057 508:148 508:5A8 566:44E 587:057 587:297 635:45F 644:43F 645:43F 647:43F)
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312:020))))))
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396:010 435:766) (44 2:405 509:417))
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214:272 (54 10:564 308:448)) (56 44:658 52:458 (54 233:274 314:767 314:347)))
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142:788 693:30B)
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(15 5:015 (38 96:014 (14 5:645 (17 308:628 348:555 (16 11:013 75:222 103:417) (16 11:013 17:224
90:727))))))
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374:022))))))
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E Complete reduced-envelope certificate

This is the complete decimal certificate of Appendix A. It begins at card 6 and ends at card 2015, after the six bases specified there. A marker beginning with # states the next card number; it is not itself a card. Each other nonempty logical line is one composition. Every three-digit block, including its leading zeros, belongs to the decimal grammar. For readable line breaking, the displayed numeric rows have a bracketed card number and spaces between blocks. Remove that bracketed prefix and those spaces to recover each exact numeric line of the literal file; markers and blank lines are unchanged. This reversible formatting is checked in the source distribution. Long rows may wrap visually.

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F Complete letter-and-offset certificate

This listing supplies every row 6 through 615 of the 35-move certificate. Appendix B defines the six base cards, every letter and translation offset, and the complete reconstruction. Each line starts with its row number, followed by a nonempty list of placed earlier cards. Long logical lines may wrap visually.

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6 0a03
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11 6e25 6g23
12 0g54 0a03
13 6g03 6e05
14 0g54 3g14
15 0g64 3g24
16 3g34 0g74
17 0g54 3a33
18 3g24 3a43
19 6e15 6a65
20 6e05 6a55
21 4g23 6a65
22 0g75 3a54
23 0g64 0a13
24 0a03 17a54
25 0a03 14a54

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28 6b45 0g64
29 1e73 6d45
30 0a13 14a64
31 0a14 17a65
32 0e83 6d45
33 0a24 17a75
34 6b45 1g54
35 0a24 14a75
36 0a34 17a85
37 6d45 2e63
38 3a33 16a24
39 0a13 17a64
40 3a43 16a34
41 6d35 1e63
42 0g64 6e15
43 2e53 6d35
44 4e15 7a45
45 6e05 7a45
46 6g03 7a45
47 6d35 6c73
48 6d45 6c83
49 4h31 7b54
50 6d35 7b54
51 10c53 10a54 26c34 26a33
52 0a03 0e73
53 6e34 3b45
54 22a23 3g14
55 7e05 7a45
56 7e05 7g03
57 0g54 7a45
58 0e54 7a45
59 0g54 7e05
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62 6g03 53a15
63 4d43 50a54
64 6g34 49b85
65 6g03 49b54
66 6d45 53f64
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69 6d45 34a65
70 4d43 41a54
71 6b45 23c43
72 0f38 8g54
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75 0a13 9a73
76 0h57 14a75
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79 6b55 48a34
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81 4d43 43a54
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83 0h57 67c63
84 6d45 67a64
85 9a63 8e54
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88 0g64 67d25
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92 6b45 33a23
93 4d43 47a54

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96 6b55 70a35
97 6g03 44a54
98 6g23 44a74
99 2b32 24a54
100 0g54 48a14
101 0h46 24a54
102 6d45 25c63
103 6d45 71a64
104 6a54 75a34
105 6f43 77a45
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107 6g03 63b53
108 4g13 63b53
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110 6g13 63b63
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113 1g64 36g75
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115 6c73 70h25
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 514 57a54 443e15 510a15
 515 2g24 512b15
 516 329a43 244e24 10b45 506e24
 517 496a23 140a42 193b42
 518 9g34 514a74
 519 498a43 311e34 38b45 408a34
 520 254a23 200f64 271g53 215c53 252e24
 521 492e51 2h42
 522 176d46 338c33 252b46
 523 303b54 25e15 107c54 511e16
 524 504e53 504a33 180a45 180e45 261a34
 525 303b54 25e15 430a46 107c54
 526 359a43 416e24 327f43 489a54 399d24
 527 387e43 38d45 491a54 519a64
 528 414b35 386b46 502f43 226e15 359f34
 529 55e34 441a43 63d33 64g75
 530 456h42 336a55 283h42 271f55
 531 425a54 336d43 340g43 394b54 423a54 526a54 401e43 453a54
 532 504e53 6h30
 533 318e45 319h51 407e41 173d45 217b45
 534 434a46 12g13
 535 444a45 426a45 25a54 321e42
 536 307a43 244a54 26d43 534a52
 537 373c24 271d33 176b46 330b46
 538 63d33 470a45 64g75 55e34
 539 51b54 488d43 428a54 516a54
 540 334a43 51b54 88d43 536a54
 541 236a55 142h41 160d41 194a53 501e12 412a53
 542 6f50 442a45
 543 493d26 107c54 318f54
 544 263c74 9g34 451b83
 545 256a45 448a54 153a63 343a45
 546 64a43 482h43 443a75 64g75
 547 443a75 510e75 131d34
 548 426a45 541a45 25a54 321e42
 549 385a64 385e24 330a64 252g24 326a64 529a24
 550 424a46 172h42 215b55 456h42
 551 300g43 172h43 319d57 527e57 176f43
 552 443e15 64c15 504b33 404f23
 553 63a35 261e54 471a51 442a45 504a33 180a45
 554 30g54 96h45 474a45
 555 504f35 513a15 55a54 443e15 471b15
 556 34b42 549a34
 557 460a54 460e24 460c54 411e54 531a36 531e46 406f54
 558 180f54 443e15 513a15 442b54
 559 228e52 7d34 155e45 538f45
 560 517f54 416f43 359f34 356g34 483a46
 561 295a54 173c54 263g14 236b55 546e54 176c54 535a45
 562 6h30 559a45
 563 167a74 538a34 514a74 330e34 545a55
 564 560a43 421a34 401b34 340d34 509a34 528a43 423f43 425f43 336a34
 565 443e45 131d34 538a34 459a55 144e34
 566 314a34 180a45 442a45 466a63 504a33
 567 21b43 561a64 500e31
 568 538e64 547a24 275a64 565a24 330a64
 569 34d42 568e54


```

570 569a25 2h43
571 293a43 570a43 382a43 556e24
572 564b66 465d34 385e34 364h34 499b65
573 527c31 499e34 527g51 499a54 407e41 531h24 407a41
574 497a24 539e46 355a24 390d24 544a24 462b45 563e54 540e46 518a24 495e24
575 314e54 504e53 180e45 442e45 559a45
576 495a54 544e54 462d35 540a36 355e54 539a36 497e54 390b54 563a24 518e54
577 64h54 64b31 443b54 443h31
578 574e45 2b43
579 2b43 576e45
580 2h42 579a33
581 578e53 2f42
582 40d43 446f53 523a70 480e53 567e54
583 25f43 525a54 582a28 221d37
584 263b45 236e35 548d34 295d45 577a45 173b45
585 236c51 300e45 479b45 577a45 183d45 527g51
586 465e45 558b45 385b45 379e45 524a45 469c45 467c45 527e55 553a45 552b45 555b45 564a64 583a52
587 300c41 479b45 319d55 477g41 252d45 467c45
588 3b33 586a45
589 4b43 588a45
590 295h43 173h43 577a47 490a47 217f43
591 524a45 529b41 529d41 538b41 538d41 197f41
592 588a45 108h30
593 2h42 586a45
594 30b43 476b43 588a45
595 460e24 460c54 586b54 411e54 551b24 406f54 572a24
596 385f42 592a56 330b46 529f46 385b46 330f42
597 24g23 221a64 199d24 594b65
598 594a56 456f42 254b32 554b32 465a46
599 578e53 492a31 512e31 586e45 485a53 579e53
600 448a45 278c14 592f53 403f54 146f43
601 437g41 553a45 592a55 236e35 469e41
602 553a47 467e43 592a57 437g43 236e37
603 505b54 505f54 586f54 551h54 531g42 586b54 585b54 585f54 551d54 573b54 533f54 533b54
604 597a34 375c43 597b43 375d34 350d43 350c34
605 400f43 383g34 530e43 596b34 522c43 598a43 537d34 520b43 550e43
606 123b43 274a43 605a64 119b43
607 420d12 419d14 589a55 314a34 571b73
608 486e45 515b45 580a45 521a45 581e45 593a45 599e45 606b62
609 286e45 107d45 532e45 542a45 221d45 562e45 575e45 604b64
610 221c54 286f54 107c54 472b54 566b54 532f54 542b54 609a26
611 456c54 456e14 594h13 594b55
612 589b55 589h03
613 607f54 503d14 592h13 592f53 607h14 612e14 612a54
614 608b75 543d56 370g34 392h34 610d56 600a64 613a64
615 602h64 614b43 533f64 590h64 611g34 587d34 601f64 406b64 591d34 586d34 584f64 406d34

```

G Complete executable verifier

The verifier below reads the literal certificate and evaluates the set operations described in the paper. From the paper root, run

```
python3 verification/verify_certificate.py
```

An explicit input path may be supplied with `--table`. The wrapper rejects any data bytes with a different checksum and refuses optimized Python execution, which would disable the assertions in the checking function. Successful termination means that every executed check has passed; the semantic implication is the induction in Proposition 5 and Lemma 3.

The arrays `AA`, `TT`, and `HH` hold the required sets, envelopes, and heights of completed numbered cards. The recursive function `expr` evaluates an inline expression in the current line's coordinate frame. A reference uses an earlier array entry; it never interprets an inline pivot as a translation.

All computations use exact Python integers and finite sets.

```
def verify(data):
    symbols = '0123456789ABCDEFG'
    snake = [(0,0),(1,0),(2,0),(3,0),(3,1),(4,1)]
    TT = [set(snake) for _ in snake]
    AA = [t - {p} for t, p in zip(TT, snake)]
    HH = [1 for _ in snake]
    def point(x):
        assert len(x) == 2
        return tuple(symbols.index(c) for c in x)
    def reference(token):
        parts = token.split(':')
        j = int(parts[0])
        assert 0 <= j < len(AA)
        if len(parts) == 1:
            s = dx = dy = 0
        else:
            assert len(parts) == 2 and len(parts[1]) == 3
            s = int(parts[1][0])
            assert 0 <= s < 8
            dx, dy = point(parts[1][1:])
    def put(P):
        Q = set()
        for x, y in P:
            if s & 2:
                x = -x
            if s & 4:
                y = -y
            if s & 1:
                x, y = y, x
            Q.add((x + dx, y + dy))
        return Q
    return put(AA[j]), put(TT[j]), HH[j]
def expr(it):
    p = point(next(it))
    aa, tt, hh = [], [], []
    while True:
        word = next(it)
        if word == ')':
            break
        a, t, h = expr(it) if word == '(' else reference(word)
        aa.append(a)
        tt.append(t)
        hh.append(h)
    assert aa
    return ((set.union(*aa) | set.intersection(*tt)) - {p},
            set.union({p}, *tt), 1 + max(hh))
for line in data.strip().splitlines():
    words = line.replace('(', ' ( ').replace(')', ' ) ').split()
    it = iter(words + [''])
    assert int(next(it)) == len(AA)
    a, t, h = expr(it)
    assert a <= t
    assert next(it, None) is None
    AA.append(a)
    TT.append(t)
    HH.append(h)
assert len(AA) == 728 and len(TT[-1]) == 251
assert all(0 <= x <= 16 and 0 <= y <= 16 for x, y in TT[-1])
assert AA[-1] == set()
assert HH[-1] == 21
# Additional checks for the stated intermediate table and examples.
assert max(HH) == 21
assert AA[6] == {(0, y) for y in range(5)}
assert TT[6] == AA[6] | {(1, 3), (1, 4), (1, 5)}
assert AA[7] == {(0, y) for y in range(1, 5)}
assert words[:2] == ['727', '88']
```

```

final_children = words[2:]
assert len(final_children) == 32
groups = [
    (648, (4, 5), 87, 15, 4, 35),
    (708, (8, 8), 225, 20, 4, 31),
    (712, (5, 5), 96, 15, 8, 21),
    (713, (5, 5), 98, 15, 4, 12),
    (725, (7, 7), 167, 20, 8, 4),
    (726, (7, 7), 169, 20, 4, 0),
]
used, common = 0, None
for j, required_cell, size, height, count, remainder in groups:
    assert AA[j] == {required_cell}
    assert len(TT[j]) == size and HH[j] == height
    for token in final_children[used:used + count]:
        assert int(token.split(':')[0]) == j
        a, t, h = reference(token)
        assert a == {(8, 8)}
        common = t if common is None else common & t
    assert len(common - {(8, 8)}) == remainder
    used += count
assert used == len(final_children)
assert common == {(8, 8)}

def main():
    import argparse
    import hashlib
    import json
    from pathlib import Path
    import sys

    if not __debug__:
        raise SystemExit("Verification requires assertions; do not use python -O.")
    parser = argparse.ArgumentParser(description="Check the exact 21-turn Snaky certificate.")
    parser.add_argument("--table", type=Path,
                        default=Path(__file__).with_name("certificate.txt"))
    args = parser.parse_args()
    payload = args.table.read_bytes()
    digest = hashlib.sha256(payload).hexdigest()
    expected = "3fa12d36a6d4dbb185e3f2808c8dfde13d85ee309afbca030d6ad17ae9fe3d04"
    if digest != expected:
        raise SystemExit("Certificate SHA256 mismatch: " + digest)
    data = payload.decode("ascii")
    verify(data)
    print(json.dumps({"status": "PASS", "certificate_sha256": digest,
                     "numbered_cards": 728, "final_index": 727,
                     "final_required_cells": 0, "final_support_cells": 251,
                     "final_height": 21}, sort_keys=True))

if __name__ == "__main__":
    main()

```

G.1 Independent reconstructions of all three certificates

The source distribution contains an additional, independently parsed reconstruction for every dialect, together with tests of the finite identities used in the article. From the paper root run

```
python3 verification/supporting/verify_all.py --output-dir ../all-checks
```

Choose a new output directory outside the paper directory. The command runs the primary and independent checks for all three literal tables and records their full-card agreement, exact identities, and rejection tests. The support guide gives each program's standalone command. The 21-move

programs under `verification/supporting/route21/` additionally check every inline node and local reply class; they are distinct from the short verifier printed above. The 25-move programs under `verification/supporting/route25/` use the reduced sets and swap-first anchor maps. The programs under `verification/supporting/route35/` check the letter-and-offset table and its conditional and tactical identities. All three literal tables and every checking program are included in the editable source distribution. The directory `verification/supporting/prior-35/` contains only a copy of the 35-move literal certificate. The scope of the finite reconstruction supplement is stated in Appendix C.